

Qatar-9b

R-2505

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-9_250111 · created 2026-08-24T23:46:54+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2460686.8466 +/- 0.0037 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.151 +/- 0.015
Transit depth (Rp/Rs)^2	2.28 +/- 0.44 [%]
Orbital Inclination (inc)	86.08 +/- 1.4
Airmass coefficient 1 (a1)	1.193 +/- 0.025
Airmass coefficient 2 (a2)	-0.147 +/- 0.041
Scatter in the residuals of the lightcurve fit is	2.1 %
Best Comparison Star	#3 - [227, 238]
Optimal Method	PSF photometry
Transit Duration (day)	0.0696 +/- 0.006

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.151 ± 0.015 vs 0.1489 (deviation 0.09σ)
Duration fit vs published	0.0696 d vs 0.0779 d (deviation 0.91σ)
Mid-time O-C	18.0 min
Depth SNR	6.6
Residual scatter	2.1 %

Light curve

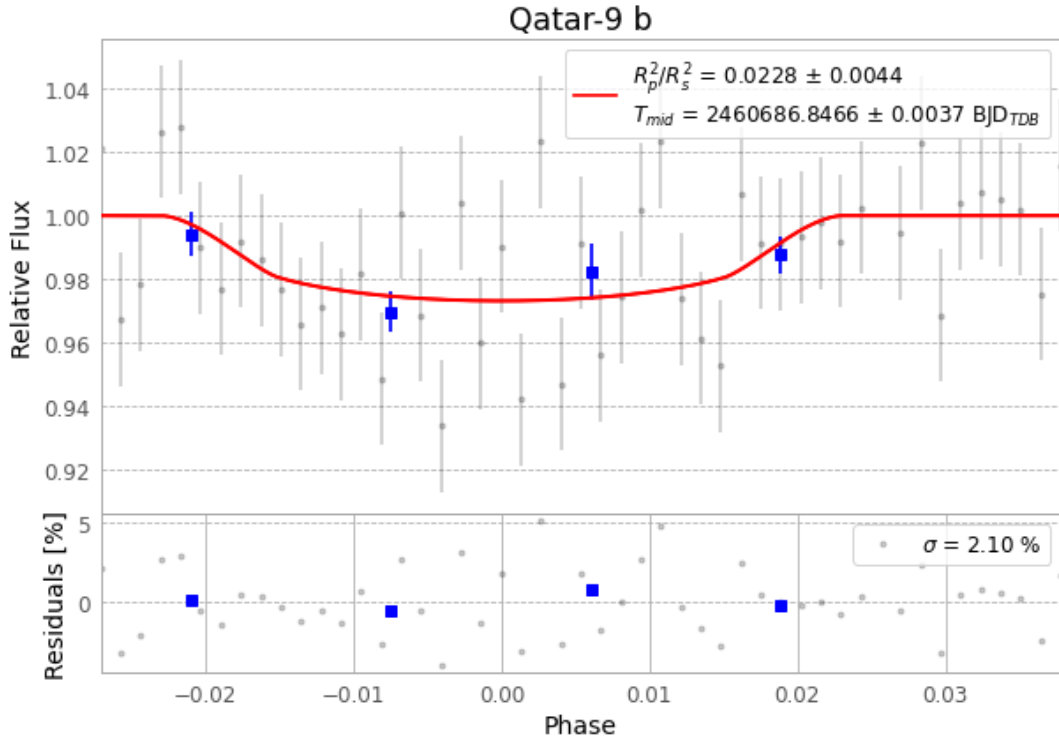


Figure 1 — FinalLightCurve_Qatar-9 b_2025-01-11.png (SHA-256 099584e02474f08c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [468, 206], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 4b56384c0f2ebffa...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

48 FITS frames (32 MB), Qatar-9250111071215.FITS → Qatar-9250111093616.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-9-250111.log (cb356bbb3e7c...), FinalLightCurve_Qatar-9 b_2025-01-11.png (099584e02474...), FinalLightCurve_Qatar-9 b_2025-01-11.csv (712703517c6e...)

Compute resources

Wall time 87.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-24 23:46Z.